Estimation and imputation in PPCA models with Missing Not At Random Data

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Motivations

- Missing Not At Random data (MNAR) type are very common in practice but they have not been much considered, as they require modeling the missing value mechanism [1].
- Very few theoretical guarantees already exist for informative missing data. Recent results have been proposed by Pearl and Mohan [2] based on graphical models and causality to handle the MNAR data in the linear models.
- **Goal:** to perform a PPCA model [3] containing several MNAR variables and to propose some consistent estimation results.

Setting

Consider

- a data matrix $Y \in \mathbb{R}^{n \times p}$,
- a coefficients matrix $B \in \mathbb{R}^{r \times p}$ of rank $r < \min\{n, p\}$,
- r latent variables grouped in $W \in \mathbb{R}^{n \times r}$,
- $M \in \mathbb{R}^{n \times p}$ the missing-data pattern: $M_{ij} = \begin{cases} 1 \text{ if } Y_{ij} \text{ is observed,} \\ 0 \text{ otherwise.} \end{cases}$

$$Y = \begin{array}{c} \begin{array}{cccccc} Y_{.1} & Y_{.2} & Y_{.3} & Y_{.4} & \dots & Y_{.p} \end{array} \\ \begin{pmatrix} 12 & 28 & 31 & 3 & \dots & \text{NA} \\ \text{NA} & 23 & 89 & 2 & \dots & 85 \\ 32 & 6 & 24 & \text{NA} & \dots & \text{NA} \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ \text{NA} & 3 & 7 & \text{NA} & \dots & 11 \end{pmatrix} \end{array}$$

Probabilistic Principal Component Analysis model

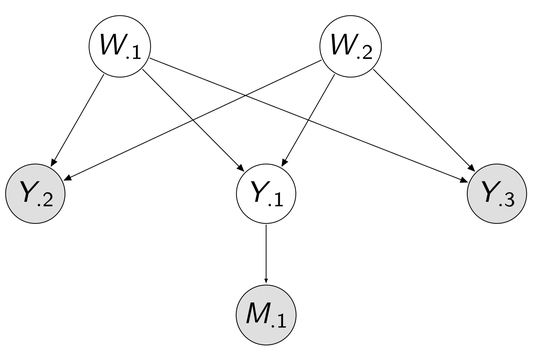
$$Y = \mathbf{1}\alpha + WB + \epsilon, \text{with} \begin{cases} W = (W_{1.}|\dots|W_{n.})^T, \text{ with } W_{i.} \sim \mathcal{N}(0_r, \mathrm{Id}_{r\times r}), \\ \alpha \in \mathbb{R}^p \text{ and } \mathbf{1} = (1\dots1)^T \in \mathbb{R}^n, \\ \epsilon = (\epsilon_{1.}|\dots|\epsilon_{n.})^T, \text{ with } \epsilon_{i.} \sim \mathcal{N}(0_p, \sigma^2 \mathrm{Id}_{p\times p}). \end{cases}$$

Proposal

Under the PPCA model:

- ✓ Mean, variance and covariances estimation of the missing variables **without explicitly modeling the MNAR mechanism and by only using observed data** $\rightsquigarrow$ **consistency results**,
- ✓ Estimation of the loading matrix B,
- ✓ Imputation of the missing values in the data matrix Y.

Toy example for $p = 3$, $r = 2$ and one missing variable $Y_{.1}$



$$(Y_{.1}\ Y_{.2}\ Y_{.3}) = \mathbf{1}\,(\alpha_1\ \alpha_2\ \alpha_3) + (W_{.1}\ W_{.2})\,B + \epsilon.$$

Idea: exploiting linear links btw variables, provided by the latent variable structure

$$Y_{.2} = \mathcal{B}_{2\to1,3[0]} + \mathcal{B}_{2\to1,3[1]}(Y_{.1} - \epsilon_{.1}) + \mathcal{B}_{2\to1,3[3]}(Y_{.3} - \epsilon_{.3}) + \epsilon_{.2}$$

- Definition: effects of $Y_{.2}$ on $(Y_{.1}, Y_{.3})$ **in the complete case** when $M_{.1} = 1$

$$(Y_{.2}|M_{.1} = 1) := \mathcal{B}^c_{2\to1,3[0]} + \mathcal{B}^c_{2\to1,3[1]}(Y_{.1} - \epsilon_{.1}) + \mathcal{B}^c_{2\to1,3[2]}(Y_{.3} - \epsilon_{.3}) + \epsilon_{.2}.$$

Main results

Theorem: Mean estimator

Assume that $\mathcal{B}^c_{2\to1,3[1]}$ is non-zero and

- $(B_{.1}\ B_{.3})$ is an invertible matrix,
- $Y_{.2} \perp\!\!\!\perp M_{.1} | Y_{.1}, Y_{.3}$ ($\rightsquigarrow$ ok for the self-masked MNAR mechanism),

$$\hat{\alpha}_1 := \frac{\hat{\alpha}_2 - \hat{\mathcal{B}}^c_{2\to1,3[0]} - \hat{\mathcal{B}}^c_{2\to1,3[3]}\hat{\alpha}_3}{\hat{\mathcal{B}}^c_{2\to1,3[1]}}.$$

▸ **Consistency** if there exist consistent estimators for α_2, α_3 and $(\mathcal{B}^c_{2\to1,3[k]})_{k=0,1,2}$.

▸ Same methodology for the variance and covariances.

▸ **Performing PPCA:** using the covariance matrix estimator

$$\hat{\Sigma} = \begin{pmatrix} \widehat{\mathrm{Var}}(Y_{.1}) & \widehat{\mathrm{Cov}}(Y_{.1}, Y_{.2}) & \widehat{\mathrm{Cov}}(Y_{.1}, Y_{.3}) \\ \widehat{\mathrm{Cov}}(Y_{.2}, Y_{.1}) & \widehat{\mathrm{Var}}(Y_{.2}) & \widehat{\mathrm{Cov}}(Y_{.2}, Y_{.3}) \\ \widehat{\mathrm{Cov}}(Y_{.3}, Y_{.1}) & \widehat{\mathrm{Cov}}(Y_{.3}, Y_{.2}) & \widehat{\mathrm{Var}}(Y_{.3}) \end{pmatrix}$$

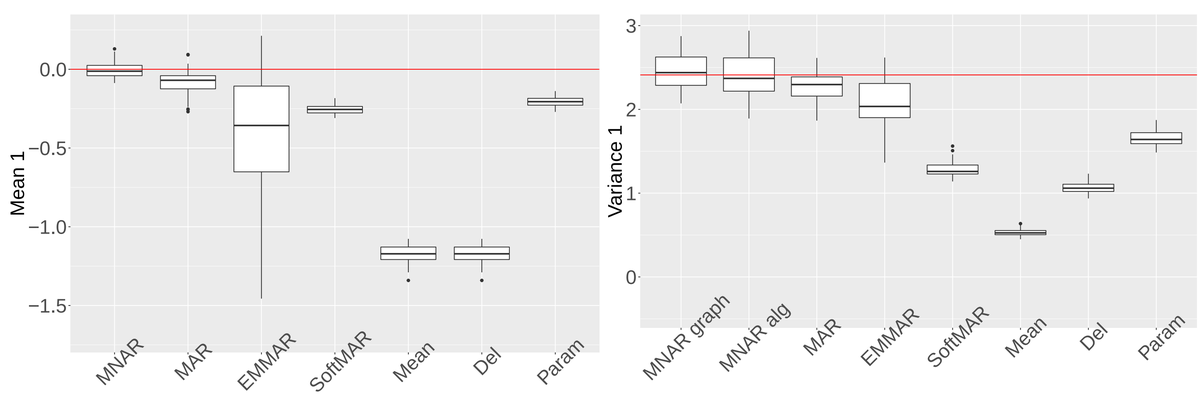
and its singular value decomposition $\hat{\Sigma} - \sigma^2 \mathrm{Id}_{3\times3} =: \hat{U}\hat{D}\hat{U}^T$, with $\hat{U} = (\hat{u}_1|\hat{u}_2|\hat{u}_3)$,

$$\hat{B} := \hat{D}^{1/2}_{|2}\hat{U}^T_{|2} = \begin{pmatrix} \sqrt{\hat{d}_1} & 0 \\ 0 & \sqrt{\hat{d}_2} \end{pmatrix} \begin{pmatrix} \hat{u}_1^T \\ \hat{u}_2^T \end{pmatrix}.$$

▸ **Imputation:** using the conditional expectation given the observable variables, Y_{i1} is imputed by

$$\hat{Y}_{i1} = \hat{\alpha}_1 + (\hat{A}_{12}\ \hat{A}_{13}) \begin{pmatrix} \hat{A}_{22} & \hat{A}_{23} \\ \hat{A}_{32} & \hat{A}_{33} \end{pmatrix}^{-1} \left(\begin{pmatrix} Y_{i2} \\ Y_{i3} \end{pmatrix} - \begin{pmatrix} \hat{\alpha}_2 \\ \hat{\alpha}_3 \end{pmatrix} \right), \text{ with } \hat{A} = \hat{B}^T\hat{B} + \sigma^2 \mathrm{Id}_{3\times3}.$$

Simulation study when $r = 2$, $n = 1000$, $p = 10$, $\sigma = 0.1$ and 7 variables are missing leading to 35 % of MNAR values.



Considering MNAR data:

- MNAR graph & alg: the proposed method,
- Param: modeling the mechanism in low rank model with fixed effects [4].

Considering MAR data:

- MAR: extension to the PPCA model of the method in [2],
- EMMAR: classical method using an EM algorithm [5],
- SoftMAR: considering a low rank model with fixed effects [6],
- Del: the listwise deletion method, by using the fully observed rows only,
- Mean: the naive imputation by the mean.

	RV coefficient	MSE
MNAR graph	0.997	0.122
MNAR alg	0.997	0.122
Param		0.337
MAR	0.997	0.152
EMMAR	0.935	0.286
SoftMAR	0.935	0.440
Del	0.772	1.57
Mean	0.726	1.63

References

[1] Roderick JA Little and Donald B Rubin. *Statistical analysis with missing data*, volume 333. John Wiley & Sons, 2014.

[2] Karthika Mohan, Felix Thoemmes, and Judea Pearl. Estimation with incomplete data: The linear case. In *IJCAI*, pages 5082–5088, 2018.

[3] Michael E Tipping and Christopher M Bishop. Probabilistic principal component analysis. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 61(3):611–622, 1999.

[4] Aude Sportisse, Claire Boyer, and Julie Josse. Imputation and low-rank estimation with missing non at random data. *arXiv preprint arXiv:1812.11409*, 2018.

[5] Alexander Ilin and Tapani Raiko. Practical approaches to principal component analysis in the presence of missing values. *Journal of Machine Learning Research*, 11(Jul):1957–2000, 2010.

[6] Trevor Hastie and Rahul Mazumder. *softImpute: Matrix Completion via Iterative Soft-Thresholded SVD*, 2015. R package version 1.4.